SOME IMPORTANT FORMULAE:

Utilization of sender Us=Ws/(1+2(dp/dt))
Ws=sender window
dp=propagation delay
dt=transmission delay

Ws(max)=1+2(dp/dt)

	STOP & WAIT	**GO BACK-N**	**SELECTIVE REPEAT**
UTILIZATION	1/(1+2(dp/dt))	N/(1+2(dp/dt)) (N=Ws)	N/(1+2(dp/dt)) (N=Ws)
SEQUENCE NUMBER(Ws+Wr)	1(Ws)+1(Wr)=2	N(Ws)+1(Wr)=N+1	N(Ws)+N(Wr)=2N
RETRANSMISSION	1	N	1

Sequence Number=Ws+Wr
No. of bits for sequence number=log2(Ws+Wr)

QUESTIONS:

1)Using stop and wait we want to send 10 packets out of which every 4th packet is lost. How many total packets needs to be sent?

Solution:

Loss Loss Loss

→ 1 2 3 ~~4~~ 4 5 6 ~~7~~ 7 8 9 ~~10~~ 10

→ Total packet sent = 13

2)If we want to send 400 packets over a channel having error probability of 0.2, then how many total packets need to be sent using stop and wait protocol to complete communication?

Solution:

$$400 + 400(0.2) + 400(0.2)^2 + 400(0.2)^3 + \cdots$$

$$= n + np + np^2 + \cdots \Rightarrow n\left(\frac{1}{1-p}\right)$$

$$= n(1 + p + p^2 + \cdots) = \frac{n}{1-p} = \frac{400}{1-0.2} = \underline{\underline{500}}$$

3)In Go Back-3 if the 5th packet being transmitted is lost and we want to send 10 packets how many transmissions are needed?

Solution:

Loss   loss

1 2 3 4 $\boxed{5\ 6\ 7}$ 5 6 $\boxed{7\ 8\ 9}$ 7 8 $\boxed{9\ 10}$ 9 10

Total packet = $\underline{\underline{18}}$

4)Solve question 3 for selective repeat.

Solution:

loss   loss

1 2 3 4 5 5 6 7 8 9 9 10

Total Transmissim = 12

5)Sender wants to achieve 60% utilization. The transmission and propagation delay are 1 msec and 49.5 msec respective. How many bits are needed to represent sequence numbers if the protocol used is selective repeat?

Solution:

$$U_s = \frac{W_s}{1+2(dp/dt)}$$

$$0.6 = \frac{W_s}{1+2(49.5/1)}$$

(60%)

$$0.6 = \frac{W_s}{1+99} = \frac{W_s}{100}$$

$$W_s = 0.6 \times 100$$

$$\boxed{W_s = 60}$$

As it is Selective Repeat.

$$\boxed{W_R = W_s}$$

START WRITING HEAR
(Begin answer for each question on a new page)

$$\boxed{W_R = 60}$$

~~No. of Sequence No be~~
~~Seq~~

Sequence No required $= W_s + W_R$

$= 60 + 60$

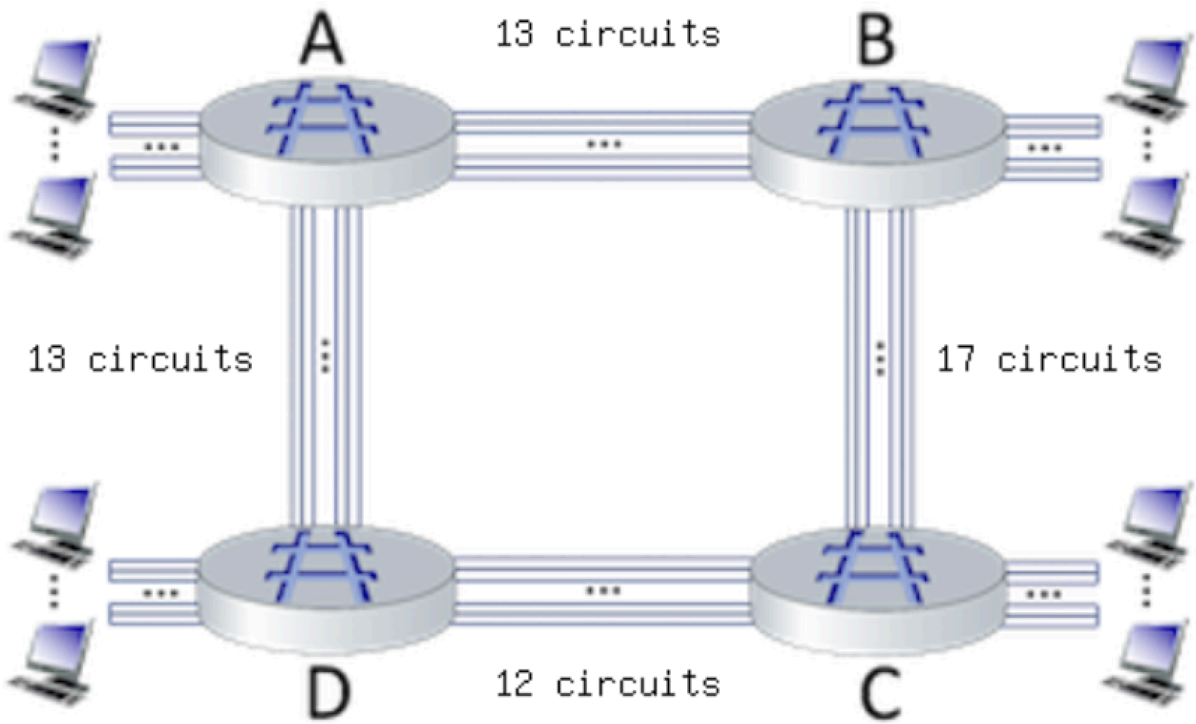
$= \boxed{120}$

No. of bit need for sequence Numbe $= \log_2(W_R + W_s)$

$= \log_2(120)$

$\approx \boxed{7 \text{ bit}}$

6)Consider the circuit-switched network shown in the figure below, with circuit switches A, B, C, and D. Suppose there are 13 circuits between A and B, 17 circuits between B and C, 12 circuits between C and D, and 13 circuits between D and A.



a)What is the maximum number of connections that can be ongoing in the network at any one time?

Solution:
The maximum number of connections is the sum of all links:
13+17+12+13=55

b)Suppose that these maximum number of connections are all ongoing. What happens when another call connection request arrives to the network, will it be accepted? Answer Yes or No

Solution:
If all connections are full, the request will be blocked:
So NO

c)Suppose that every connection requires 2 consecutive hops, and calls are connected clockwise. For example, a connection can go from A to C, from B to D, from C to A, and from D to B. With these constraints, what is the maximum number of connections that can be ongoing in the network at any one time?

Solution:

- A → C (A → B → C) is limited by the smallest capacity in the path:

    - A → B has 13 circuits
    - B → C has 17 circuits
    - The bottleneck is A → B (13 circuits)
    - So, at most 13 connections can be A → C at once.
- C → A (C → D → A) is limited by:
    - C → D has 12 circuits
    - D → A has 13 circuits
    - The bottleneck is C → D (12 circuits)
    - So, at most 12 connections can be C → A at once.

Total max connections from A → C and C → A = 13 + 12 = 25 connections.

B → D (B → C → D) is limited by:

    - B → C has 17 circuits
    - C → D has 12 circuits
    - Bottleneck is C → D (12 circuits)
    - So, max 12 connections for B → D.
- D → B (D → A → B) is limited by:
    - D → A has 13 circuits
    - A → B has 13 circuits
    - Bottleneck is A → B (13 circuits)
    - So, max 13 connections for D → B.

Total max connections from B → D and D → B = 12 + 13 = 25 connections.

- The total sum of max possible connections from A → C and C → A = 25.
- The total sum of max possible connections from B → D and D → B = 25.

Since we can't have more than 25 ongoing connections at once due to circuit limitations, the final maximum number of ongoing connections at any given time is 25.

Ans:25

d)Suppose that 19 connections are needed from A to C, and 16 connections are needed from B to D. Can we route these calls through the four links to accommodate all 35 connections? Answer Yes or No.

Solution:
The sum of our needed connections is 35, and we have 25 available connections, so it is NOT possible.

7)A circuit-switching scenario in which N_{cs} users, each requiring a bandwidth of 25 Mbps, must share a link of capacity 200 Mbps.
A packet-switching scenario with N_{ps} users sharing a 200 Mbps link, where each user again requires 25 Mbps when transmitting, but only needs to transmit 30 percent of the time

a)When circuit switching is used, what is the maximum number of users that can be supported?

Solution:
When circuit switching is used, at most 8 users can be supported. This is because each circuit-switched user must be allocated its 25 Mbps bandwidth, and there is 200 Mbps of link capacity that can be allocated.

b)Suppose packet switching is used. If there are 15 packet-switching users, can this many users be supported under circuit-switching? Yes or No.

Solution:
No. Under circuit switching, the 15 users would each need to be allocated 25 Mbps, for an aggregate of 375 Mbps - more than the 200 Mbps of link capacity available.

c)Suppose packet switching is used. What is the probability that a given (specific) user is transmitting, and the remaining users are not transmitting?(Nps=15)

Solution:
The probability that a given (specific) user is busy transmitting= p, is just the fraction of time it is transmitting, i.e. 0.3.
The probability that one specific other user is not busy is (1-p),
So the probability that all of the other Nps-1 users are not transmitting is (1-p)^(Nps-1).
Thus the probability that one specific user is transmitting and the remaining users are not transmitting is p*(1-p)^(Nps-1),
=0.3*(0.7)^(15-1)
=0.002

d)Suppose packet switching is used. What is the probability that one user (any one among the 15 users) is transmitting, and the remaining users are not transmitting?

Solution:
The probability that exactly one (any one) of the Nps users is transmitting is Nps times the probability that a given specific user is transmitting and the remaining users are not transmitting.
The answer is thus Nps * p * (1-p)^(Nps-1)
=15*0.002
=0.03

e)When one user is transmitting, what fraction of the link capacity will be used by this user? Write your answer as a decimal.

Solution:
This user will be transmitting at a rate of 25 Mbps over the 200 Mbps link.
=25/200=0.125=0.13.

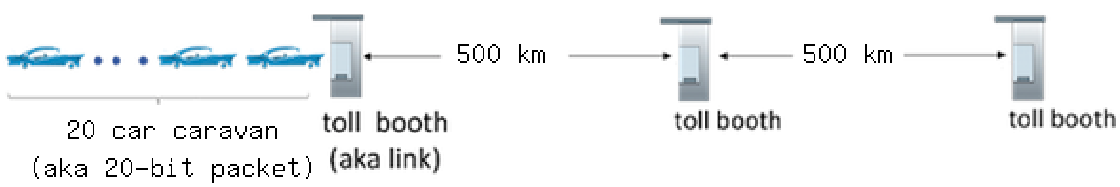
f)What is the probability that any 5 users (of the total 15 users) are transmitting and the remaining users are not transmitting?

Solution:
The probability that 5 specific users of the total 15 users are transmitting and the other 10 users are idle is p^5(1-p)^10.
Thus the probability that any 5 of the 15 users are busy is choose(15, 5) * p^5(1-p)^10, where choose(15, 5) is the (15, 5) coefficient of the binomial distribution.
=15c5*(0.3)^5*(0.7)^10
=3003*0.00243*0.0282
=0.206
=0.21

g)What is the probability that more than 8 users are transmitting?

Solution:
The probability that more than 8 users of the total 15 users are transmitting
= Σ i=9,15 choose(15, i) * p^i*(1-p)^(15 - i)
=0.015.

8)Consider the figure below, adapted from Figure 1.17 in the text, which draws the analogy between store-and-forward link transmission and propagation of bits in packet along a link, and cars in a caravan being serviced at a toll booth and then driving along a road to the next tollbooth.



Suppose the caravan has 20 cars, and that the tollbooth services (that is, transmits) a car at a rate of one car per 2 seconds. Once receiving serving a car proceeds to the next tool both, which is 500 kilometers away at a rate of 20 kilometers per second. Also assume that whenever the first car of the caravan arrives at a tollbooth, it must wait at the entrance to the tollbooth until all of the other cars in its caravan have arrived, and lined up behind it before being serviced at the toll booth. (That is, the entire caravan must be stored at the tollbooth before the first car in the caravan can pay its toll and begin driving towards the next tollbooth).

a)Once a car enters service at the tollbooth, how long does it take until it leaves service?
Solution:
Service time is 2 seconds

b)How long does it take for the entire caravan to receive service at the tollbooth (that is the time from when the first car enters service until the last car leaves the tollbooth)?

Solution:
It takes 40 seconds to service every car, (20 cars * 2 seconds per car)

c)Once the first car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?

Solution:
500km/(20 km/sec)
=25 seconds

d)Once the last car leaves the tollbooth, how long does it take until it arrives at the next tollbooth?

Solution:
Just like in the previous question, it takes 25 seconds, regardless of the car

e)Once the first car leaves the tollbooth, how long does it take until it enters service at the next tollbooth?

Solution:
(20-1) cars *2 seconds per car + 500km/(20 km/sec)
=19*2+25
=63 seconds

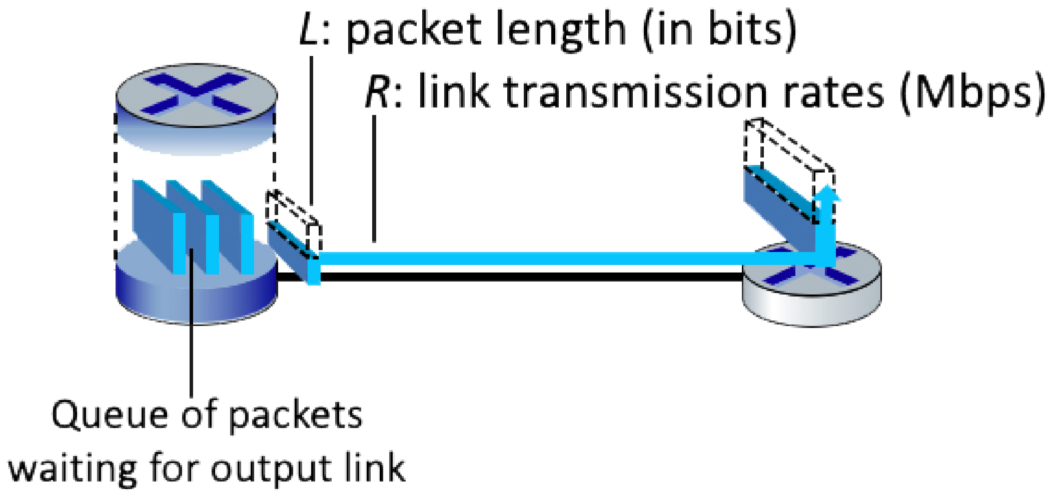
f)Are there ever two cars in service at the same time, one at the first toll booth and one at the second toll booth? Answer Yes or No

Solution:
No, because cars can't get service at the next tollbooth until all cars have arrived

g)Are there ever zero cars in service at the same time, i.e., the caravan of cars has finished at the first toll booth but not yet arrived at the second tollbooth? Answer Yes or No

Solution:
Yes, one notable example is when the last car in the caravan is serviced but is still travelling to the next toll booth; all other cars have to wait until it arrives, thus no cars are being serviced

9)Consider the figure below, in which a single router is transmitting packets, each of length *L* bits, over a single link with transmission rate *R* Mbps to another router at the other end of the link.



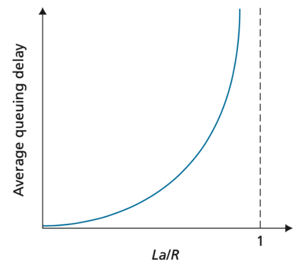
Suppose that the packet length is *L*= 4000 bits, and that the link transmission rate along the link to router on the right is *R* = 1000 Mbps.

Round your answer to two decimals after leading zeros

a)What is the transmission delay?
Solution:
The transmission delay = L/R
 = 4000 bits / 1000000000 bps
 = 4*10^(-6) seconds

b)What is the maximum number of packets per second that can be transmitted by this link?
Solution:
The number of packets that can be transmitted in a second into the link
= R / L
= 1000000000 bps / 4000 bits
= 250000 packets

10)Consider the queuing delay in a router buffer, where the packet experiences a delay as it waits to be transmitted onto the link. The length of the queuing delay of a specific packet will depend on the number of earlier-arriving packets that are queued and waiting for transmission onto the link. If the queue is empty and no other packet is currently being transmitted, then our packet's queuing delay will be zero. On the other hand, if the traffic is heavy and many other packets are also waiting to be transmitted, the queuing delay will be long.



Assume a constant transmission rate of R = 2000000 bps, a constant packet-length L = 3500 bits, and a is the average rate of packets/second. Traffic intensity I = La/R, and the queuing delay is calculated as I(L/R)(1 - I) for I < 1.

a)In practice, does the queuing delay tend to vary a lot? Answer with Yes or No.
Solution:
Yes, in practice, queuing delay can vary significantly. We use the above formulas as a way to give a rough estimate.

b)Assuming that a = 28, what is the queuing delay? Give your answer in milliseconds (ms)
Solution:
Queuing Delay = I(L/R)(1 - I) * 1000
= (3500*28/2000000)*(3500/2000000)*(1-(3500*28/2000000)) * 1000
= 0.049*(3500/2000000)*(1-0.049)*(1000)
=0.0815ms

c)Assuming that a = 61, what is the queuing delay? Give your answer in milliseconds (ms)
Solution:
Queuing Delay = I(L/R)(1 - I) * 1000
= (3500*61/2000000)*(3500/2000000)*(1-(3500*61/2000000)) * 1000
= 0.1068*(3500/2000000)*(1-0.1068)*(1000)
=0.1669ms

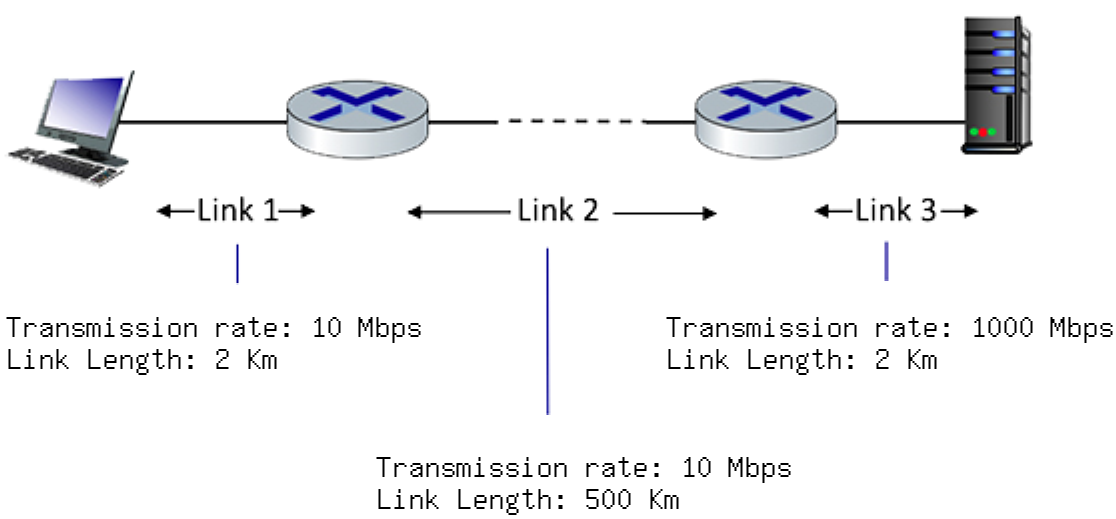
d)Assuming the router's buffer is infinite, the queuing delay is 0.1669 ms, and 1856 packets arrive. How many packets will be in the buffer 1 second later?
Solution:
Queuing delay = 0.1669 ms
→ This means that each packet spends 0.1669 ms in the queue before being processed.
Number of arriving packets initially = 1856 packets
Time duration considered = 1 second (1000 ms)
Router buffer is infinite, meaning no packet loss due to overflow.

The Number of Packets Processed in 1 Second=1000/0.1669
≈5982.03
=5982
So, the router can process 5982 packets in 1 second.

Since 5982>1856
Therefore no packets are left in the buffer.

e)If the buffer has a maximum size of 845 packets, how many of the 1856 packets would be dropped upon arrival from the previous question?
Solution:
Packets dropped = packets - buffer size
= 1856 - 845
= 1011 dropped packets.

11)Consider the figure below, with three links, each with the specified transmission rate and link length.



Assume the length of a packet is 12000 bits. The speed of light propagation delay on each link is 3x10^8 m/sec

Round your answer to two decimals after leading zeros

a) What is the transmission delay of link 1?
Solution:
Link 1 transmission delay = L/R
= 12000 bits / 10 Mbps
= 0.0012 seconds

b) What is the propagation delay of link 1?
Solution:
Link 1 propagation delay = d/s
= (2 Km) * 1000 / 3*10^8 m/sec
= 6.67*10^(-6) seconds

c) What is the total delay of link 1?
Solution:
Link 1 total delay = d_t + d_p
= 0.0012 seconds + 6.67*10^(-6) seconds
= 0.0012 seconds

d) What is the transmission delay of link 2?
Solution:
Link 2 transmission delay = L/R
= 12000 bits / 10 Mbps
= 0.0012 seconds

e) What is the propagation delay of link 2?
Solution:
Link 2 propagation delay = d/s
= (500 Km) * 1000 / 3*10^8 m/sec
= 0.0017 seconds

f) What is the total delay of link 2?
Solution:
Link 2 total delay = d_t + d_p
= 0.0012 seconds + 0.0017 seconds
= 0.0029 seconds

g) What is the transmission delay of link 3?
Solution:
Link 3 transmission delay = L/R
= 12000 bits / 1000 Mbps
= 1.20*10^(-5) seconds

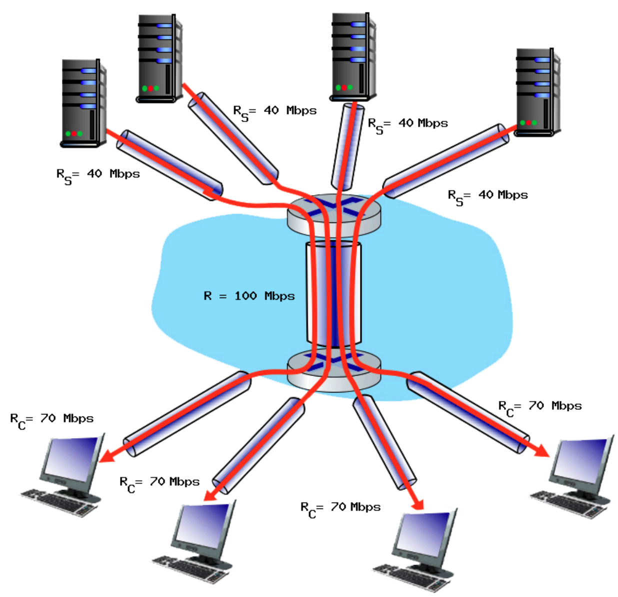
h) What is the propagation delay of link 3?
Solution:
Link 3 propagation delay = d/s
= (2 Km) * 1000 / 3*10^8 m/sec
= 6.67*10^(-6) seconds

i) What is the total delay of link 3?
Solution:

Link 3 total delay = d_t + d_p
= 1.20*10^(-5) seconds + 6.67*10^(-6) seconds
= 1.87*10^(-5) seconds

j)What is the total delay?
Solution:
The total delay = d_L1 + d_L2 + d_L3
= 0.0012 seconds + 0.0029 seconds + 1.87E-5 seconds
= 0.0041 seconds

12)Consider the scenario shown below, with four different servers connected to four different clients over four three-hop paths. The four pairs share a common middle hop with a transmission capacity of R = 100 Mbps. The four links from the servers to the shared link have a transmission capacity of R_S = 40 Mbps. Each of the four links from the shared middle link to a client has a transmission capacity of R_C = 70 Mbps.



a)What is the maximum achievable end-end throughput (in Mbps) for each of four client-to-server pairs, assuming that the middle link is fairly shared (divides its transmission rate equally)?
Solution:
The maximum achievable end-end throughput is the capacity of the link with the minimum capacity, which is 25 Mbps(min of 40,25,70)

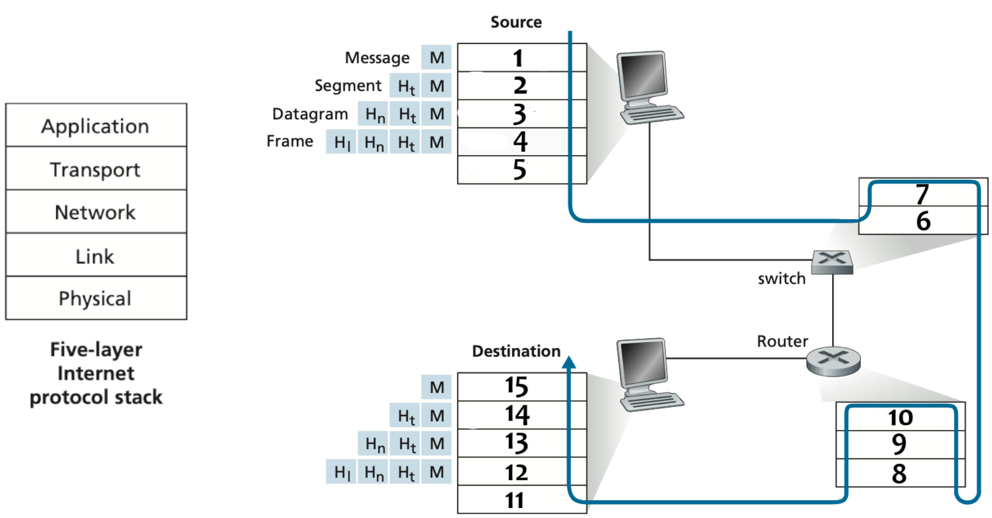
b) Which link is the bottleneck link? Format as Rc, Rs, or R
Solution:
The bottleneck link is the link with the smallest capacity between RS, RC, and R/4. The bottleneck link is R.

c) Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the server links (RS)? Answer as a decimal
Solution:
The server's utilization = Rbottleneck / RS
= 25 / 40
= 0.63

d) Assuming that the servers are sending at the maximum rate possible, what are the link utilizations for the client links (RC)? Answer as a decimal
Solution:
The client's utilization = Rbottleneck / RC
= 25 / 70
= 0.36

e) Assuming that the servers are sending at the maximum rate possible, what is the link utilizations for the shared link (R)? Answer as a decimal
Solution:
The shared link's utilization = Rbottleneck / (R / 4)
= 25 / (100 / 4)
= 1

13)In the scenario below, imagine that you're sending an http request to another machine somewhere on the network.



a)What layer in the IP stack best corresponds to the phrase: 'handles the delivery of segments from the application layer, may be reliable or unreliable'
ans:The given phrase corresponds to the Transport Layer

b) What layer in the IP stack best corresponds to the phrase: 'bits live on the wire'
ans:The given phrase corresponds to the Physical Layer

c)What layer in the IP stack best corresponds to the phrase: 'moves datagrams from the source host to the destination host'
ans:The given phrase corresponds to the Network Layer.

d) What layer in the IP stack best corresponds to the phrase: 'handles messages from a variety of network applications'
ans:The given phrase corresponds to the Application Layer.

e) What layer in the IP stack best corresponds to the phrase: 'passes frames from one node to another across some medium'
ans:The given phrase corresponds to the Link Layer.

f)What layer corresponds to box 1-15?
ans:Box 1 is the Application Layer.
Box 2 is the Transport Layer.
Box 3 is the Network Layer.
Box 4 is the Link Layer.
Box 5 is the Physical Layer.
Box 6 is the Physical Layer.
Box 7 is the Link Layer.
Box 8 is the Physical Layer.
Box 9 is the Link Layer.
Box 10 is the Network Layer.
Box 11 is the Physical Layer.
Box 12 is the Link Layer.
Box 13 is the Network Layer.
Box 14 is the Transport Layer.
Box 15 is the Application Layer.

14)Consider the two 16-bit words (shown in binary) below. Recall that to compute the Internet checksum of a set of 16-bit words, we compute the one's complement sum [1] of the two words. That is, we add the two numbers together, making sure that any carry into the 17th bit of this initial sum is added back into the 1's place of the resulting sum); we then take the one's complement of the result. Compute the Internet checksum value for these two 16-bit words:

00110000 11110001 this binary number is 12529 decimal (base 10)

01100001 00110101 this binary number is 24885 decimal (base 10)

a)What is the sum of these two 16 bit numbers? Don't put any spaces in your answer
Solution:
The sum of 00110000 11110001 and 01100001 00110101 = 10010010 00100110

b)Using the sum from question 1, what is the checksum? Don't put any spaces in your answer
Solution:
The internet checksum is the one's complement of the sum: 10010010 00100110 = 01101101 11011001
.

15)Dr. Johnson is conducting an experiment to analyze the performance of reliable data transfer protocols in the transport layer. In this experiment, the sender and receiver are located 120 kilometers apart and are connected using a long-range Wi-Fi link with a propagation speed of 3,000 meters per second. The data packet size is 2 MB, and the link has, a transmission rate of 10 Mbps. For all calculations, assume maximum utilization of the link.
i) Determine Propagation Delay.
ii) Determine Transmission Delay
iii) Determine the minimum sender window size required for the Go-Back-N protocol.
iv) After experimentation Dr. Johnson came up with a conclusion that
"Queuing delay is responsible for network congestion".
. Is this conclusion
valid? Justify your answer.

Solution:
Given Data:
Distance between sender and receiver: 120 km = 120,000 meters
Propagation speed: 3,000 m/s
Packet size: 2 MB = 2 × 10^6 × 8 bits = 16 × 10^6 bits
Transmission rate: 10 Mbps = 10 × 10^6 bps

(i) Propagation Delay
Propagation Delay=DistancePropagation /Speed
=120,000 m/3,000 m/s=
=40 seconds
So, Propagation Delay = 40 seconds

(ii) Transmission Delay
Transmission Delay=Packet Size (in bits)/Transmission Rate (in bps)
=16×10^6 bits/10×106 bps
= 1.6 sec
So, Transmission Delay = 1.6 seconds.

(iii) Minimum Sender Window Size for Go-Back-N (GBN) Protocol:
Utilization=Ws/(1+2(dp/dt))
Since utilization is 1
Therefore
Ws= (1+2(dp/dt))
=1+2(40/1.6)
=51
(iv) Is Dr. Johnson's Conclusion Valid?
Dr. Johnson concluded that "Queuing delay is responsible for network congestion."
This is partially correct, but queuing delay is not the only cause of congestion.
Justification:
Queuing delay occurs when packets wait in a buffer before transmission.
Network congestion can also be caused by:
Insufficient bandwidth (if demand exceeds capacity).
High traffic load from multiple sources.
Propagation delay in long-distance networks.
Packet loss and retransmissions due to errors.
Unoptimized routing causing congestion at specific nodes.
Thus, while queuing delay contributes to congestion, it is not the sole reason.

16)Consider an instance of TCP's Additive Increase Multiplicative Decrease (AIMD) algorithm where the window size at the start of the slow start phase is 2 MSS and the threshold at the start of the first transmission is 8 MSS. Assume that a timeout occurs during the fifth transmission. Find the congestion window size at the end of the tenth transmission.

Solution:
Given:

- Initial congestion window size (CWND) = 2 MSS
- Threshold (SSTHRESH) = 8 MSS
- Timeout occurs during the fifth transmission
- We need to find the congestion window size at the end of the tenth transmission

Step 1: Slow Start Phase (Exponential Growth)

TCP starts in slow start, where the congestion window doubles every round-trip time (RTT) until it reaches the threshold.

Transmission	CWND (MSS)
1st	2
2nd	4
3rd	8 (Reached THRESH)
4th	9(Enters congestion avoidance)
5th	10 Timeout occurs

Before the fifth transmission, CWND was 10 MSS.

Step 2: Timeout Event (Multiplicative Decrease)

On timeout, THRESH is set to half of CWND before timeout: SSTHRESH=10/2=5MSS THRESH

CWND is reset to 2 MSS.

Step 3: Restarting from Slow Start

After a timeout, TCP restarts from 2 MSS and grows exponentially until it reaches the new THRESH (5 MSS), then switches to additive increase.Since 4 MSS is still below SSTHRESH (5 MSS), TCP remains in slow start.

Transmission	CWND (MSS)
6th	2
7th	4

Step 4: Entering Congestion Avoidance

In the 8th transmission, CWND would increase to 8 MSS if slow start continued, but since THRESH = 5 MSS, TCP enters congestion avoidance at 4 MSS.In congestion avoidance, CWND increases additively (by 1 MSS per RTT).

Transmission	CWND (MSS)

8th	5 (Enters congestion avoidance)
9th	6
10th	7

Final Answer:

At the end of the tenth transmission, the congestion window size is 7 MSS.